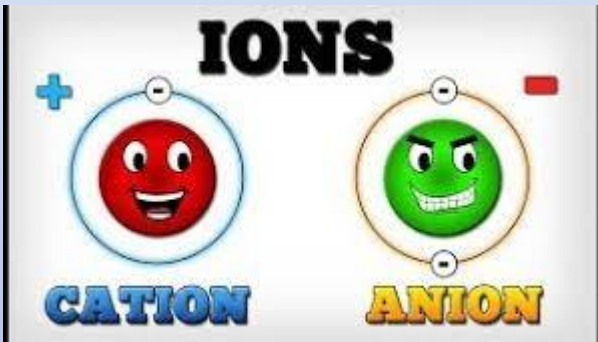


Pharmaceutical Analytical Chemistry I

Course Code (PA 102 & CPA 102)



Anion

Cation

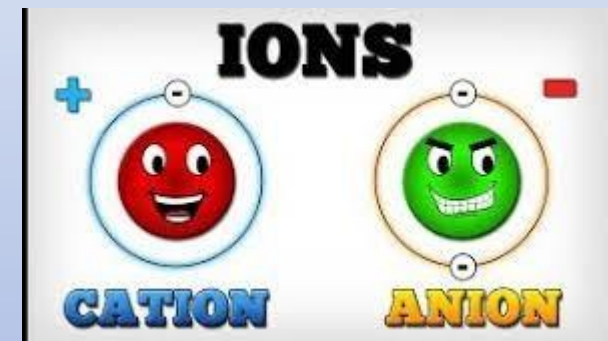
Lecture (4) Halides Containing Anions Group

Dr. Maha Mahmoud

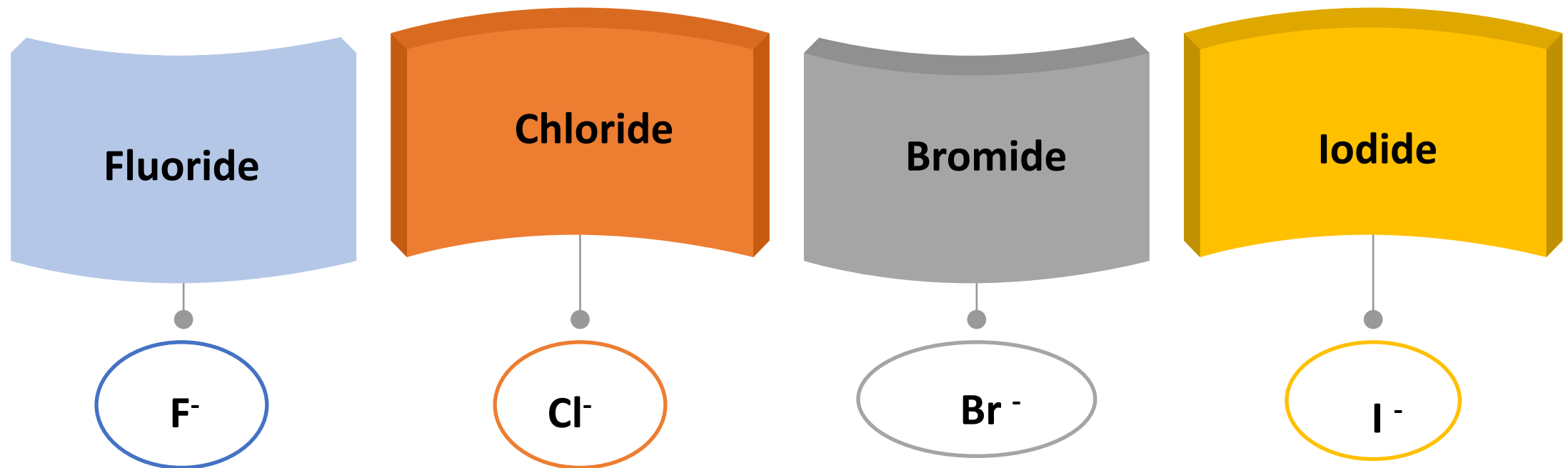
**Lecturer in Pharmaceutical Analytical
Chemistry Department, MTI University**

By the End of This Lecture , You should Know:

- 1- Halides (Members, Symbol & Charge)
- 2- Dry Reaction of Halides
- 3- Wet Reaction of Halides
- 4- Confirmatory Test of Halides
- 5- How to Analyze Mixture of Halide



Halides Containing Anions



F⁻ Fluoride

Cl⁻ Chloride

Br⁻ Bromide

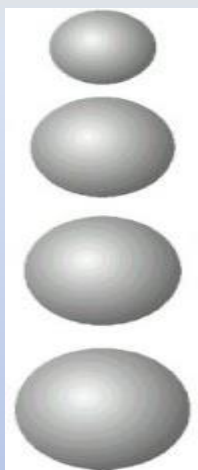
I⁻ Iodide

**Halogens Belong to gp VII
in the Periodic Table**

1	2											3	4	5	6	7	0	
		H																He
Li	Be											B	C	N	O	F	Ne	
Na	Mg											Al	Si	P	S	Cl	Ar	
K	Ca	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn	Ga	Ge	As	Se	Br	Kr	
Rb	Sr	Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd	In	Sn	Sb	Te	I	Xe	
Cs	Ba	La	Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg	Tl	Pb	Bi	Po	At	Rn	
Fr	Ra	Ac																

Group 7 Halogens

Halides are Characterized by their higher electronegativity (High tendency for gaining electrons)



F⁻

Cl⁻

Br⁻

I⁻



***Electronegativity decrease**

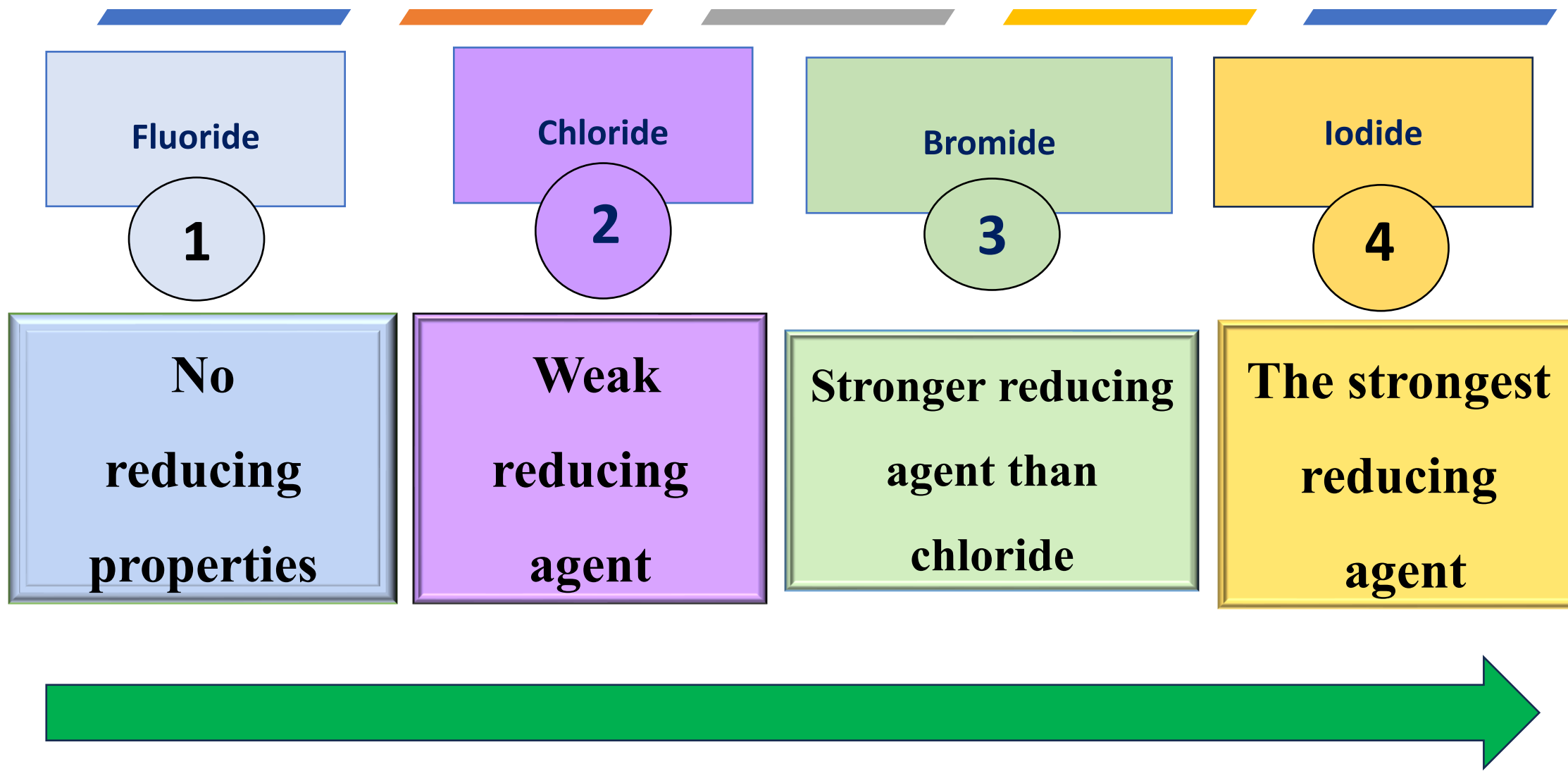
***Atomic size increase**

- As the atomic size increases, the tendency to lose electrons increases (Gain Electrons decreased).
- Iodide ion is firstly & easily oxidized into free I₂ by losing an electron followed by Br⁻ when present in a mixture.
- It's difficult to oxidize F⁻ into F₂, hence F⁻ ions are highly stable to held strongly a proton.

Therefore, the order of stronger halogen acid is from



Reducing Properties



Parent Acids

Hydrofluoric acid
HF

1

Hydrochloric acid
HCl

2

3. Hydrobromic acid
HBr

3

Hydroiodic acid
HI

4

Weakest
Acid

Strong
Acid

Stronger
than HCl

The strongest
acid



A)- Dry Reaction

1- Action of dil. HCl

2- Action of conc. H_2SO_4



1)- Action of Diluted Hydrochloric Acid



**-ve results for all
Halides**

2)- Action of Concentrated Sulphuric Acid

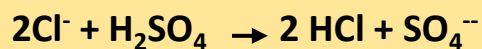
Conc.
 H_2SO_4

A)- F^-
Evolution of colorless gas (HF)

Conc.
 H_2SO_4

B)- Cl^-

HCl gas is evolved which can be identified by



i) Changing a blue moistened litmus paper into red.

ii) Forming white clouds of ammonium chloride, a glass rod moistened with ammonia



Conc.
 H_2SO_4

C)- Br^-

- HBr & Br_2 is formed which have characteristic brown color especially on warming.



Conc.
 H_2SO_4

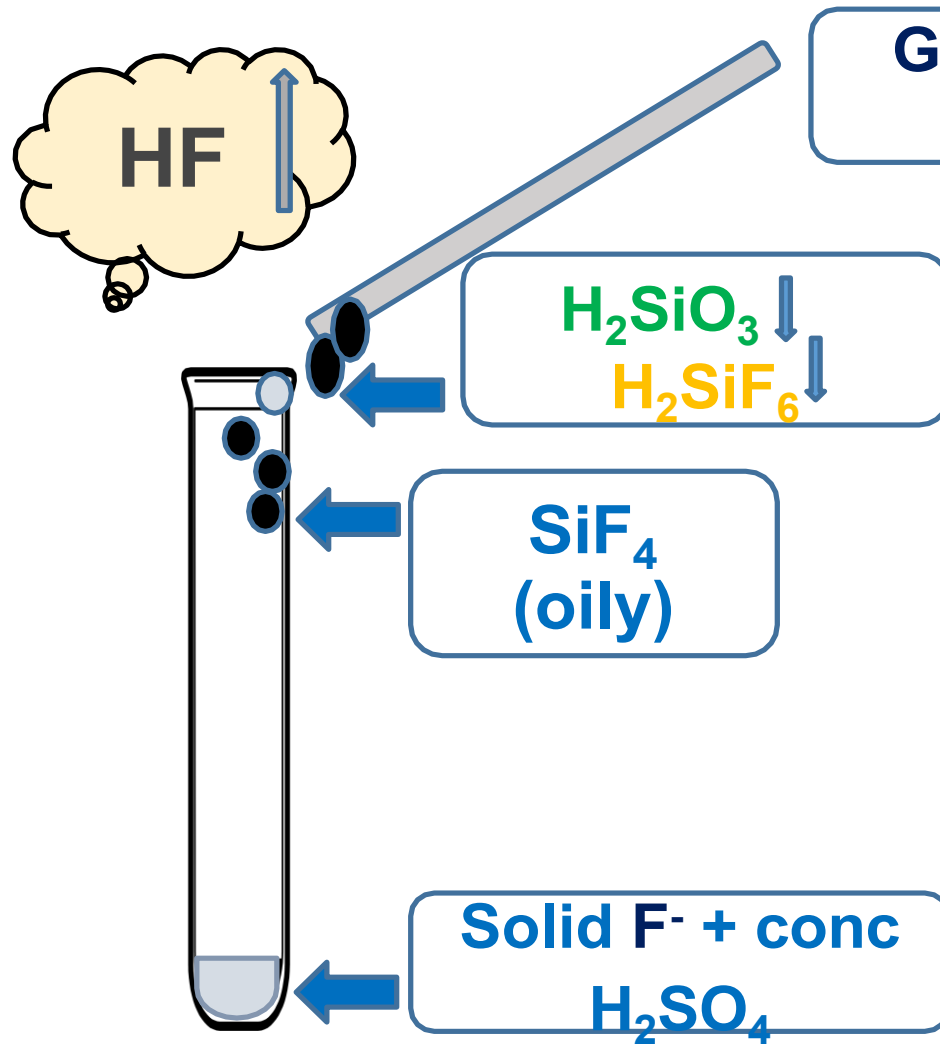
D)- I^-

- Since HI & I_2 is formed which have characteristic violet fumes.

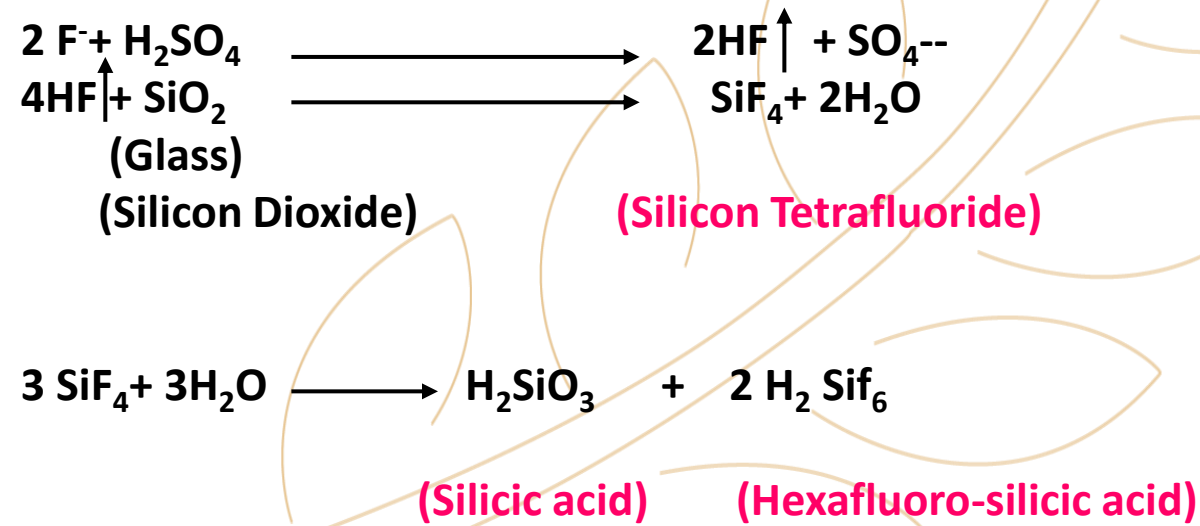


Iodine can be detected by exposing the evolved gas to paper moistened with starch solution, it changes into blue.





HF gas has Corrosive & Itching Action on glass in presence of water
The test tube gets an oily appearance & if exposed to a glass rod moistened with water, it gets white gelatinous appearance



B)-Wet Reaction



B)- Wet Reaction

1- Effect of
Silver Nitrate

3- Effect of Lead
Acetate

2- Effect of Barium
Chloride

Solubility

- All the salts of Cl^- , Br^- & I^- are soluble except Ag^+ , Hg_2^{2+} & Cu^+ salts, their lead salts are slightly soluble in cold water, soluble in hot water.

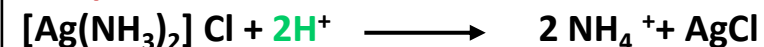
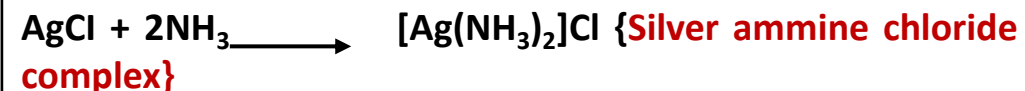
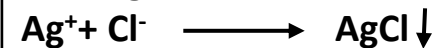
1)- Reaction with Silver Nitrate



No precipitate, since AgF is soluble in water



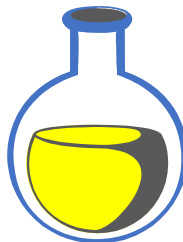
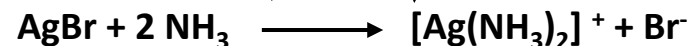
- White ppt. of AgCl is formed
- The precipitated AgCl is soluble in dil. ammonia solution to give the ammine complex.



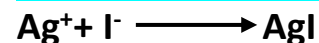
AgCl is reprecipitated upon treatment of the ammine complex with acid.



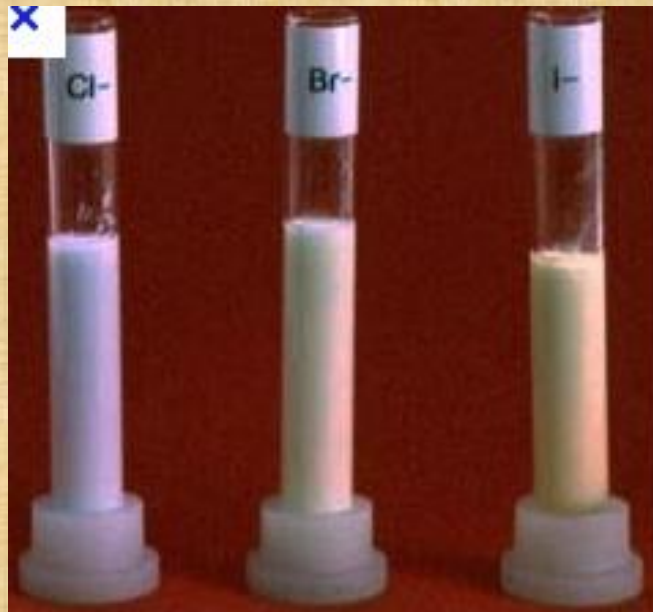
- Yellowish white precipitate of AgBr soluble in conc. ammonia solution



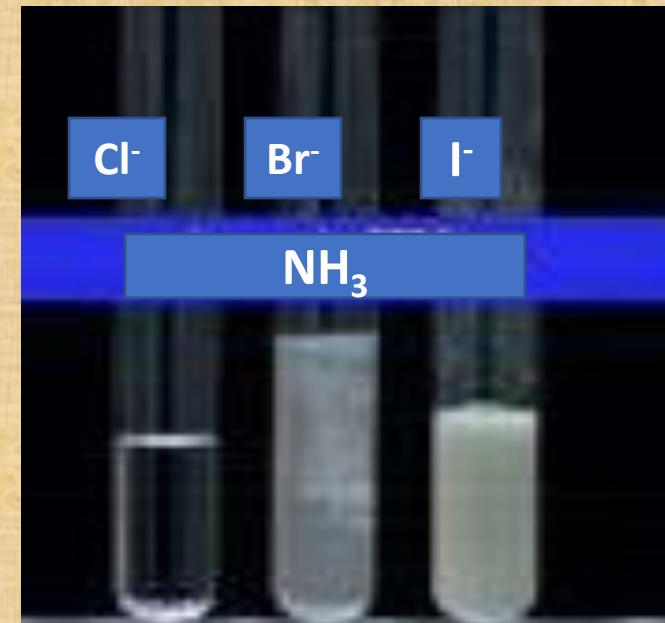
- Yellow ppt. of AgI is formed which is insoluble in dil. ammonia but very slightly soluble in conc. ammonia solution.



Reaction of Halides with Silver Nitrate



Solubility with
Diluted
ammonia

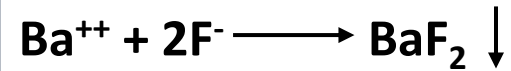


- AgI is the most insoluble one, followed by AgBr & AgCl.
- AgCl will be dissolved in dil. ammonia, followed by AgBr in conc. ammonia solution but AgI does not.
- That is due to high solubility product of AgCl (More soluble)
- AgI (Lower solubility product) (KSP) (More insoluble)

2)- Reaction with Barium Chloride



White gelatinous ppt. of BaF_2 (Barium fluoride) is formed



The ppt. is partially **soluble in dil. HCl or HNO_3 , insoluble in acetic acid.**

**+Ve Only With
Fluoride**

3)- Reaction with Lead Acetate

A)- F^- + Lead Acetate



White ppt of PbF_2 is formed, sparingly soluble in cold more soluble in hot water
 $Pb^{++} + 2 F^- \longrightarrow PbF_2$ (Lead Fluoride)

B)- Cl^- + Lead Acetate



White ppt of $PbCl_2$ is formed
 $Pb^{++} + 2 Cl^- \longrightarrow PbCl_2$ (Lead Chloride)

C)- Br^- + Lead Acetate



White ppt of $PbBr_2$ is formed
 $Pb^{++} + 2 Br^- \longrightarrow PbBr_2$ (Lead Bromide)

d)- I^- + Lead Acetate



$Pb^{++} + 2 I^- \longrightarrow PbI_2$ (Lead Iodide)
Bright yellow ppt of PbI_2 is formed which is soluble in hot water & crystallizes on cooling as golden spangles.

Special Tests



I- Fluoride

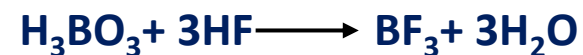
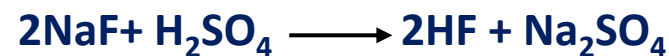
Boron Fluoride Test

When fluoride is mixed with borax & moisten with conc. H_2SO_4 . The formed HF & boric acid react to produce boron fluoride gas.

If the mixture introduced into the flame-tinged **Green** by BF_3 gas.



Boric acid



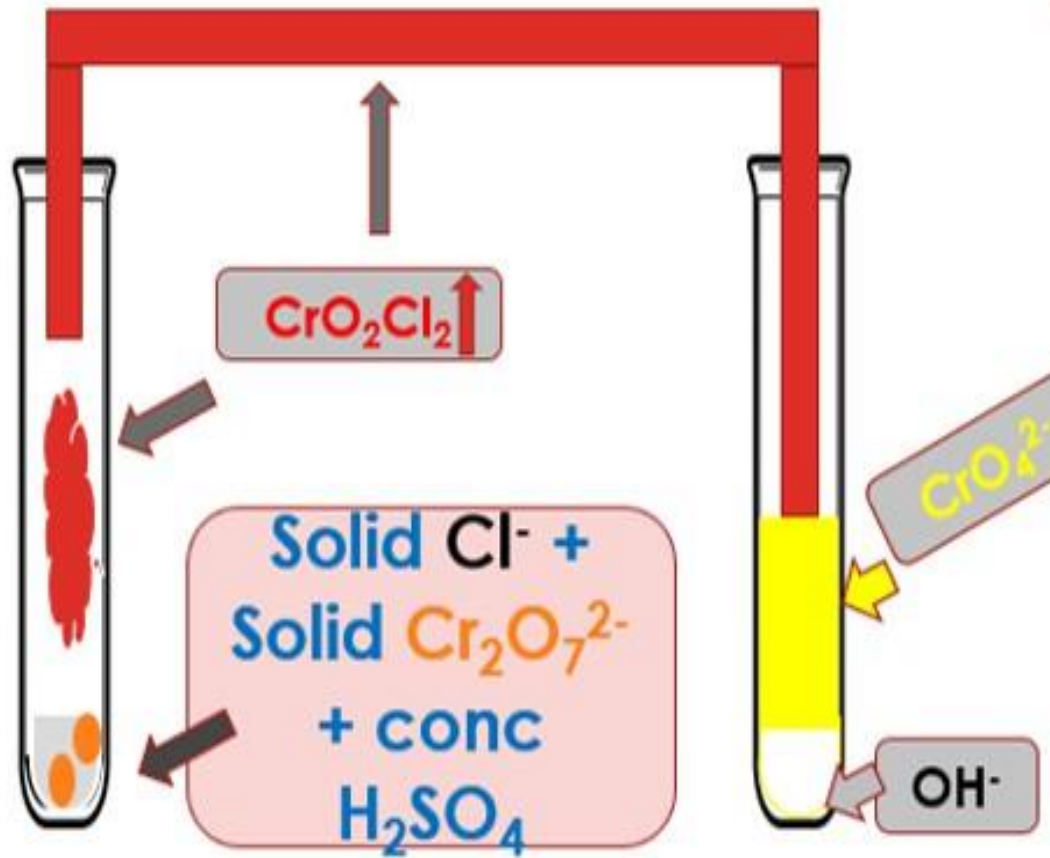
(Boron Trifluoride gas)



Special Tests:

II Chloride: Chromyl Chloride test.

One part of solid halide is added to three parts of potassium dichromate ($K_2Cr_2O_7$) and equal volume of sulphuric acid, heat gently



Dense **deep red fumes** of chromyl chloride gas (CrO_2Cl_2) are evolved:



If the fumes are passed into sodium hydroxide solution, a yellow color of chromate (CrO_4^{2-}) is obtained:



The resulting yellow solution in the test tube contains sodium chromate; this confirmed by 2 tests:



1)- Perchromic Acid Test

Acidifying the resulting solution with dil. H_2SO_4 , adding 1-2 ml ether, followed by a little H_2O_2 solution. The organic layer is colored blue



2)- Lead Acetate Test

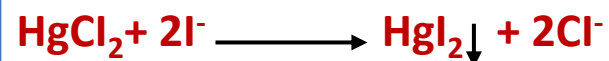
Acidifying the resulting solution with acetic acid, adding 1-2 ml lead acetate. Yellow ppt of (PbCrO_4) occurred.



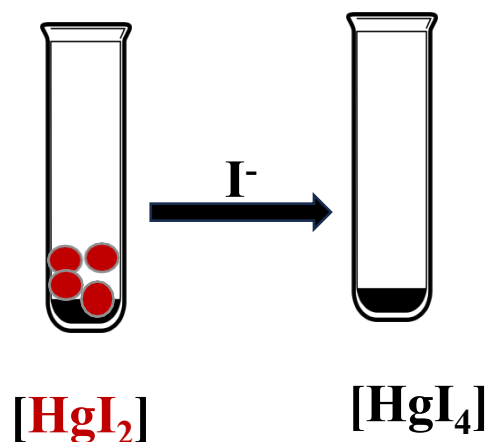
III- Iodide

Mercuric Chloride Test

I⁻ reacts with mercuric chloride solution mercuric iodide HgI₂ will be precipitated as **yellow - scarlet red** ppt. which dissolves in excess iodide forming soluble colourless complex.

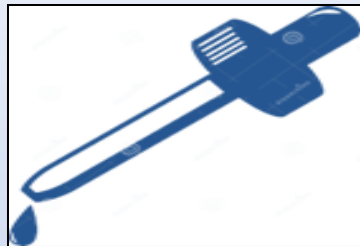


Yellow Scarlet red



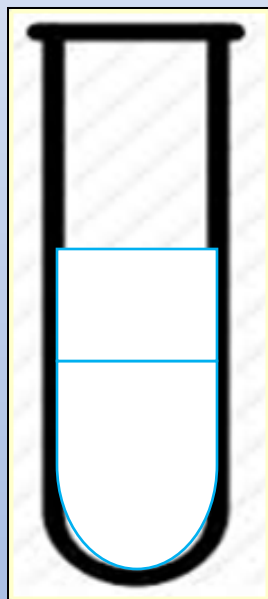
Chlorine Water Test for Bromide & Iodide

D) 1-2 dps
chlorine water



Br^- gives Brown color
in chloroform Br_2

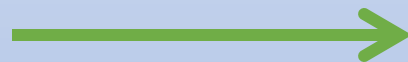
C) 2ml CHCl_3



A) 1 ml S.S

B) 2 ml dil H_2SO_4

**Vigorous
shaking**

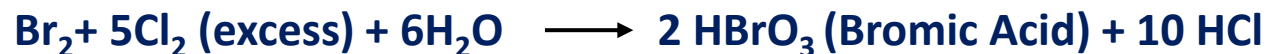
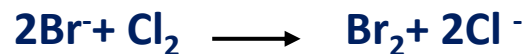
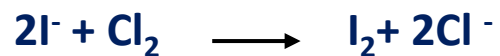


I^- gives Violet color in
chloroform I_2



Chlorine Water Test

- When chlorine water reagent is added dropwise to a solution of iodide or bromide, iodine is liberated as a brown solution, followed by bromine.
- Both I_2 and Br_2 liberated can be extracted with chloroform or carbon tetrachloride as **violet colour** followed by **brown** or yellow color of I_2 and Br_2 , respectively. **Chlorine water oxidizes I^- & Br^- into violet & brown colour in the organic layer.**
- Excess chlorine water converts Br_2 into yellow bromine monochloride or into colourless hypobromous acid or bromic acid and the organic layer turns pale yellow or colourless.

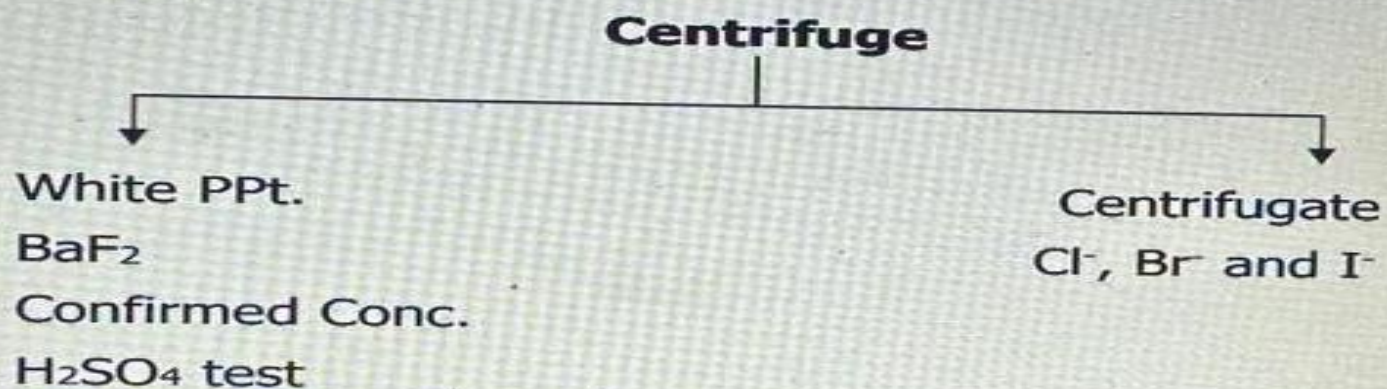


Chloride and Fluoride do not interfere with chlorine water test.

Analysis of Mixtures

1- Mixture of F^- , Cl^- , Br^- & I^-

- i. The F^- is separated by treating the mixture solution acidified with CH_3COOH with $Ba(NO_3)_2$ or $Ca(NO_3)_2$



- ii. for the centrifugate, carry out chlorine water test for both I^- and Br^-

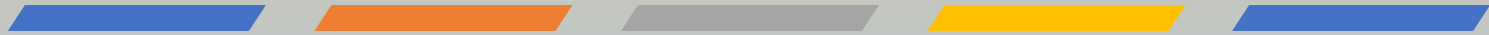
c) For Cl^- , carry out chromyl test on a solid sample.

It's possible to test Br^- after removal of I^- by:

i) oxidation in dil. H_2SO_4 by NO_2^- or H_2O_2 which oxidizes I^- into I_2 , leaving clear colourless I_2 free solution.

ii) Iodide; tested and removed by precipitation as Cu_2I_2 (by $CuSO_4$ in acid solution with dil. H_2SO_4 in presence of sulphite). The excess Cu^{2+} in the centrifugate is removed by adding $NaOH$ and boiling, centrifuge and reject the $(OH)_2$ ppt. Test the clear solution for Br^- .

Summary



Test	F -	Cl -	Br -	I -
Dry Reactions				
1- Solid + dil. HCl	- NO observed Reaction	- NO observed Reaction	- NO observed Reaction	Violet fumes of I ₂ + HI (Test by Starch paper)
2- Conc. H ₂ SO ₄	Itching of glass and oily appearance	Colorless gas (HCl), fumes with moist air, white clouds with NH ₄ OH	Brown solution, on warming, brown fumes of (Br ₂ & HBr)	
3- Chromyl chloride test. Solid + K ₂ Cr ₂ O ₇ + H ₂ SO ₄ equal volume heat gently pass into NaOH solution	Brown fumes of CrO ₂ F ₂ may be evolved	Deep red fumes of CrO ₂ Cl ₂ , which give a yellow colour to NaOH solution and confirmed by per chromic acid test or lead acetate solution	Brown vapors of Br ₂ , pale yellow in NaOH solution, negative test for CrO ₄	Violet fumes of I ₂ , pale yellow in NaOH, negative test for Cl ₂
Wet Reactions				
1- AgNO ₃ Reagent	No ppt.	White ppt., insol. in dil HNO ₃ , sol. in NH ₄ OH, KCN, S ₂ O ₃ solutions	Yellowish white ppt., AgBr, insol. In dil. HNO ₃ , slightly sol. in dil. NH ₄ OH, sol. in conc. KCN & S ₂ O ₃ solution	Yellow ppt. (AgI) insol. in conc. HNO ₃ , dil. NH ₄ OH, partially sol. in conc, NH ₄ OH. Sol. in KCN, and S ₂ O ₃ solutions
2- BaCl ₂ Reagent	White gel. ppt., of BaF ₂ , in sol. acetic acid, sol. in dil. HCl & HNO ₃	No Ppt.		
3-Lead Acetate Reagent	White ppt. PbF ₂ sol. in hot water	White ppts., soluble in hot water (PbCl ₂ , PbBr ₂) on cooling crystallize in needles		Bright yellow ppt., sol. in hot water, and crystallize on Cooling
4- Chlorine Water Test	No Reaction		Brown colour in the CHCl ₃ layer with excess Cl ₂ water changes into pale yellow (BrCl) or colourless	Violet colour in the CHCl ₃ , discharge with excess Cl ₂ water (HO ₃)

Short Exams





Anion from Halides has strong Reducing Properties



State one of the halides has high KSP



**Which of the following is the strongest
Acid**

- 1)- HF**
- 2)- HCl**
- 3)- HI**
- 4)- HBr**



**Boron Fluoride is a special test
for.....**



**Barium Chloride Gives +ve only
with**



**Conc. Sulfuric with Iodide
gives.....**



**Mercuric Chloride is a special test
for**



**Chromyl Chloride test is a special
test for.....**





